

To create a Python Application

```
Terminal

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\VICTUS> mkdir docker_python

Directory: C:\Users\VICTUS

Mode                LastWriteTime         Length Name
----                -
d-----          03-05-2026   02:49 PM             docker_python

PS C:\Users\VICTUS> cd docker_python
PS C:\Users\VICTUS\docker_python> notepad app.py
PS C:\Users\VICTUS\docker_python> cat app.py
from flask import Flask
app = Flask(__name__)
@app.route("/")
return "Hello from Docker!"

RAM 2.63 GB  CPU 0.17%  Disk: 21.19 GB used (limit 1006.85 GB)
```

Step1: create a folder

Step2: create a file name app.py

from flask import Flask

app = Flask(__name__)

@app.route("/") def

home():

return "Hello from Docker!" app.run(host="0.0.0.0",

port=5000)

```
Terminal

PS C:\Users\VICTUS\docker_python> cat requirements.txt
flask
PS C:\Users\VICTUS\docker_python> notepad dockerfile
PS C:\Users\VICTUS\docker_python> ren dockerfile.txt dockerfile
PS C:\Users\VICTUS\docker_python> cat dockerfile
# 1. Base image
FROM python:3.11-slim
# 2. Set working directory
WORKDIR /app
# 3. Copy dependency file
COPY requirements.txt
.# 4. Install dependencies
RUN pip install --no-cache-dir -r requirements.txt
# 5. Copy application code
COPY . .
# 6. Expose port
EXPOSE 5000
# 7. Run the app
CMD ["python", "app.py"]
PS C:\Users\VICTUS\docker_python>
```

Step3: create another filename requirement.txt (write flask in it)

Step4: create the dockerfile

1. Base image

FROM python:3.11-slim

2. Set working directory

WORKDIR /app

3. Copy dependency file

COPY requirements.txt

4. Install dependencies

RUN pip install --no-cache-dir -r requirements.txt

5. Copy application code

COPY . .

6. Expose port

EXPOSE 5000

7. Run the app

CMD ["python", "app.py"]

```
PS C:\Users\VICTUS\docker_python> ls

Directory: C:\Users\VICTUS\docker_python

Mode                LastWriteTime         Length Name
----                -
-a----             03-05-2026   02:50 PM             141 app.py
-a----             03-05-2026   02:52 PM             323 dockerfile
-a----             03-05-2026   02:51 PM              5 requirements.txt
```

Step5: check the directory

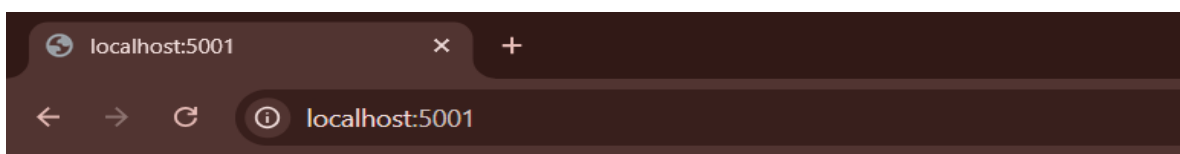
```
C:\Users\VICTUS\docker_python> docker build -t python11 .
[+] Building 16.3s (11/11) FINISHED
=> [internal] load build definition from dockerfile
=> => transferring dockerfile: 327B
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
[1/5] FROM docker.io/library/python:3.11-slim@sha256:6d85378d88a19cd4d76079817532d62232be95757cb45945a99fec8e8084b9c2
=> resolve docker.io/library/python:3.11-slim@sha256:6d85378d88a19cd4d76079817532d62232be95757cb45945a99fec8e8084b9c2
=> sha256:91ff8760033c3b9bc9c3a3bcc6ff66c113ec9e13f94fa2f12ae4b221aae996b9 1.29MB / 1.29MB
=> sha256:f3ba2250c52436c41e9f2f3830af92de466fcacc2d848264e1f9fdc76975d803 14.37MB / 14.37MB
=> sha256:7ccd73948dde4b7f1623ac6bf9b58706fb83d0a363d765d7c799d46035ceb563 249B / 249B
=> sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd 29.78MB / 29.78MB
=> extracting sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd 1.3s
=> extracting sha256:91ff8760033c3b9bc9c3a3bcc6ff66c113ec9e13f94fa2f12ae4b221aae996b9 0.2s
=> extracting sha256:f3ba2250c52436c41e9f2f3830af92de466fcacc2d848264e1f9fdc76975d803 0.8s
=> extracting sha256:7ccd73948dde4b7f1623ac6bf9b58706fb83d0a363d765d7c799d46035ceb563 0.0s
=> [internal] load build context
=> => transferring context: 597B
[2/5] WORKDIR /app
[3/5] COPY requirements.txt .
[4/5] RUN pip install --no-cache-dir -r requirements.txt
[5/5] COPY . .
exporting to image
1.2s
```

Step6: Build the image

```
S C:\Users\VICTUS\docker_python> docker run -p 5001:5000 python11
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.4:5000
press CTRL+C to quit
72.17.0.1 - - [03/May/2026 09:28:41] "GET / HTTP/1.1" 200 -
72.17.0.1 - - [03/May/2026 09:28:41] "GET /favicon.ico HTTP/1.1" 404 -
72.17.0.1 - - [03/May/2026 09:29:29] "GET / HTTP/1.1" 200 -
72.17.0.1 - - [03/May/2026 09:29:51] "GET / HTTP/1.1" 200 -
72.17.0.1 - - [03/May/2026 09:29:51] "GET /favicon.ico HTTP/1.1" 404 -
```

Step7: check the image build

Step8: do the port mapping



Step9: run the server on the browser

Create another application from python using flask

```
72.17.0.1 - - [03/May/2026 09:29:51] "GET /favicon.ico HTTP/1.1" 404 -
S C:\Users\VICTUS\docker_python> notepad app2.py
S C:\Users\VICTUS\docker_python> ren dockerfile dockerfile.txt
S C:\Users\VICTUS\docker_python> notepad dockerfile.txt
S C:\Users\VICTUS\docker_python> ren dockerfile.txt dockerfile
```

Step1: Create the new app2.py file Step2:

Write code

```
from flask import Flask, request
```

```
app = Flask(__name__)
```

```
@app.route("/", methods=["GET", "POST"]) def
```

```
home():
```

```
    if request.method == "POST":
```

```

        name = request.form.get("name")
    return f"Hello {name}! (from Docker)"

return """

<h2>Simple Flask Form</h2>

<form method="post">

    <input type="text" name="name" placeholder="Enter name" required>

    <button type="submit">Submit</button>

</form>

"""

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)

```

Step3: Again, rename the dockerfile to dockerfile.txt

Step4: Make the changes in that dockerfile.txt

1. Base image

```
FROM python:3.11-slim
```

2. Set working directory

```
WORKDIR /app
```

3. Copy dependency file

```
COPY requirements.txt
```

4. Install dependencies

```
RUN pip install --no-cache-dir -r requirements.txt
```

5. Copy application code

```
COPY . .
```

6. Expose port

```
EXPOSE 5000
```

7. Run the app

CMD ["python", "app2.py"]

Step5: Save the file and exit then rename the file from .txt to dockerfile

```
S C:\Users\VICTUS\docker_python> notepad app2.py
S C:\Users\VICTUS\docker_python> ren dockerfile dockerfile.txt
S C:\Users\VICTUS\docker_python> notepad dockerfile.txt
S C:\Users\VICTUS\docker_python> ren dockerfile.txt dockerfile
S C:\Users\VICTUS\docker_python> docker build -t python2 .
+J Building 2.2s (11/11) FINISHED                                docker:desktop-linux
=> [internal] load build definition from dockerfile                0.0s
=> => transferring dockerfile: 376B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 1.6s
=> [auth] library/python:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:6d85378d88a19cd4d76079817532d62232be95 0.0s
=> => resolve docker.io/library/python:3.11-slim@sha256:6d85378d88a19cd4d76079817532d62232be95 0.0s
=> [internal] load build context                                  0.0s
=> => transferring context: 1.00kB                                    0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> CACHED [3/5] COPY requirements.txt .                             0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> [5/5] COPY . .                                                 0.0s
=> exporting to image                                             0.3s
=> => exporting layers                                              0.1s
=> => exporting manifest sha256:7d46e1574bba64dbf5cc10d452943d0bdf5760c7aad4f17db083f4fff86672 0.0s
=> => exporting config sha256:9af5b930c8e98db96eaf2f69676697b6558724f18d5ddf1eb260b09834d141b0 0.0s
=> => naming to docker.io/library/python2:latest                  0.0s
=> => unpacking to docker.io/library/python2:latest               0.0s
```

Step6: Build the image and do the port mapping

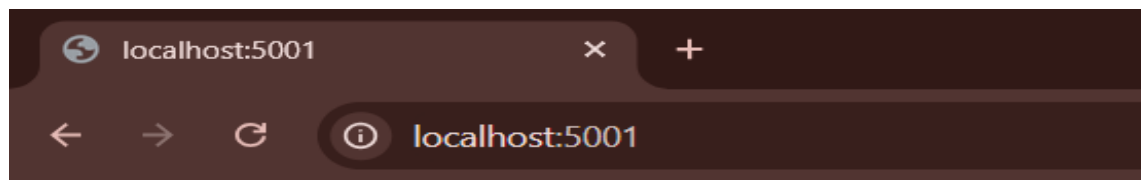
```
S C:\Users\VICTUS\docker_python> docker run -p 5001:5000 python2
* Serving Flask app 'app2'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.4:5000
press CTRL+C to quit
72.17.0.1 - - [03/May/2026 09:34:55] "GET / HTTP/1.1" 200 -
72.17.0.1 - - [03/May/2026 09:34:55] "GET /favicon.ico HTTP/1.1" 404 -
72.17.0.1 - - [03/May/2026 09:35:09] "POST / HTTP/1.1" 200 -
```

Step7: Run the server on browser



localhost:5001

Simple Flask Form



Hello Kavya Gupta! (from Docker)